

EXPERIMENT -8

Code:

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("store_inventory.csv")

# Create Pivot Table to find product-wise quantity
pivot = pd.pivot_table(
    df,
    values="Quantity",
    index="Product",
    aggfunc="sum"
)

print("Product-wise Quantity:")
print(pivot)
```

Output:

```
Product-wise Quantity:
      Quantity
Product
Keyboard      20
Laptop         8
Mouse         30
Notebook      50
Pen           65
```

EXPERIMENT -9

Code:

```
import pandas as pd

# Create the dataset
data = {
    "OrderDate": [
        "1-6-18", "1-23-18", "2-9-18", "2-26-18", "3-15-18", "4-1-18",
        "4-18-18", "5-5-18", "5-22-18", "6-8-18", "6-25-18", "7-12-18",
        "7-29-18", "8-15-18", "9-1-18", "9-18-18", "10-5-18", "10-22-18"
    ],
    "Region": [
        "East", "Central", "Central", "Central", "West", "East",
        "Central", "Central", "West", "East", "Central", "East",
        "East", "East", "Central", "East", "Central", "East"
    ],
    "Manager": [
        "Martha", "Hermann", "Hermann", "Timothy", "Timothy", "Martha",
        "Martha", "Hermann", "Douglas", "Martha", "Hermann", "Martha",
        "Douglas", "Martha", "Douglas", "Martha", "Hermann", "Martha"
    ],
    "SalesMan": [
        "Alexander", "Shellli", "Luis", "David", "Stephen", "Alexander",
        "Steven", "Luis", "Michael", "Alexander", "Sigal", "Diana",
        "Karen", "Alexander", "John", "Alexander", "Sigal", "Alexander"
    ],
    "Item": [
        "Television", "Home Theater", "Television", "Cell Phone", "Television",
        "Home Theater", "Television", "Television", "Television", "Home Theater",
        "Television", "Home Theater", "Home Theater", "Television", "Desk",
        "Video Games", "Home Theater", "Cell Phone"
    ],
    "Units": [
        95, 50, 36, 27, 56, 60, 75, 90, 32, 60, 90, 29, 81, 35, 2, 16, 28, 64
    ],
    "Unit_price": [
        1198, 500, 1198, 225, 1198, 500, 1198, 1198, 1198, 500,
        1198, 500, 500, 1198, 125, 50, 5, 500, 225
    ],
    "Sale_amt": [
        113810, 25000, 43128, 6075, 67088, 30000, 89850, 107820,
        38336, 30000, 107820, 14500, 40500, 41930, 250, 936, 14000, 14400
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

# Create Pivot Table

# Create Pivot Table
pivot = pd.pivot_table(
    df,
    values="Sale_amt",
    index=["Region", "Manager", "SalesMan"],
    aggfunc="sum"
)

print("Pivot Table:")
print(pivot)
```

Output:

```
Pivot Table:
Region Manager SalesMan Sale_amt
Central Douglas John 250
          Hermann Luis 150948
          Shellli 25000
          Sigal 121820
          Martha Steven 89850
          Timothy David 6075
East Douglas Karen 40500
      Martha Alexander 231076
          Diana 14500
West Douglas Michael 38336
      Timothy Stephen 67088
```

EXPERIMENT -10

Code:

```
import pandas as pd
import numpy as np

# Create DataFrame
np.random.seed(100)

df = pd.DataFrame({
    'A': range(1, 11),
    'B': np.random.randn(10),
    'C': np.random.randn(10),
    'D': np.random.randn(10),
    'E': np.random.randn(10)
})

# Function to highlight negative numbers
def color_negative(val):
    if val < 0:
        return 'color: red'
    else:
        return 'color: black'

# Display styled DataFrame
print(df)

# Highlight negative values in red and positive values in black
styled_df = df.style.map(color_negative)

styled_df
```

Output:

	A	B	C	D	E
0	1	-1.749765	-0.458027	1.618982	0.222400
1	2	0.342680	0.435163	1.541605	-1.443217
2	3	1.153036	-0.583595	-0.251879	-0.756352
3	4	-0.252436	0.816847	-0.842436	0.816454
4	5	0.981321	0.672721	0.184519	0.750445
5	6	0.514219	-0.104411	0.937082	-0.455947
6	7	0.221180	-0.531280	0.731000	1.189622
7	8	-1.070043	1.029733	1.361556	-1.690617
8	9	-0.189496	-0.438136	-0.326238	-1.356399
9	10	0.255001	-1.118318	0.055676	-1.232435